MIL-N-36220a
25 May 1967
SUPERSEDING
MIL-N-36220 (DM)
dated 8 June 1964

MILITARY SPECIFICATION

NEEDLE, HYPODERMIC, MALLEABLE, LEMMON, SPINAL PUNCTURE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers malleable spinal puncture needles, of the Lemmon design which are interchangeable with, and which fit the appropriate connections of the Lemmon spinal anesthesia set used in continuous spinal anesthesia.

1.2 Classification. Needles shall be of the following sizes (gages and length) (see 6.1):

Size 1 - 18 gage, 3-1/2 inch.
Size 2 - 18 gage, 4 inch.
Size 3 - 19 gage, 4 inch.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-535	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water-Resistant (for Carton Sealing).
MMM-A-250	Adhesive, Water-Resistant (For Sealing Fiberboard Boxes).

FSC 6515

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Military

MIL-P-116 Preservation, Methods of.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

20482 Anesthesia Set, Spinal, Lemmon

(Copies of specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

United States of America Standards Institute (USA Standard)

USAZ70.1-1955 (UDC615.47) American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York N.Y. 10016).

3. REQUIREMENTS

3.1 Material. The needle cannula and stylet wire shall be malleable, fabricated from class 3, alloy 752 (nickel-silver) in accordance with GG-I-526, except that the cannula and stylet wire shall not be plated. The cannula tubing shall be seamless drawn, or welded and drawn. The needle hub and stylet knob shall be fabricated from brass, nickel plated, or chromium over nickel plated in accordance with plating requirements for class 3 material of GG-I-526. As an alternate, the needle cannula and the stylet wire shall be fabricated from class 3, type 304 material in accordance with GG-I-526. The cannula shall be properly annealed so as to be malleable.

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3.2 Style, design, dimensions, and construction. The spinal puncture needles shall be malleable, of the Lemmon design, and suitable for use with the Lemmon spinal anesthesia set in continuous spinal anesthesia. Style, design, dimensions, and construction shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 20482, for appropriate stock number.

3.2.1 Cannula. The cannula shall extend into the hub sufficiently to allow for secure assembly. In accordance with standard practice, the extension of the needle cannula into the hub shall not be such as to prevent the cleaning wire from being readily inserted into the lumen of the needle cannula or to create areas for possible accumulation of materials. The cannula shall be securely attached to the hub so as to be concentric with the hub cavity.

3.2.2 Hub Shall be Luer-lock hub, so shaped as to provide means for facilitating attachment to, and removal from, a syringe. Hub and hub cavity shall be in accordance with figure 3 of USA Standard Z70.1-1955 (UDC615.47). The hubs shall be tested for diameter and depth of hub cavity as specified in 4.4.1. Hubs which do not fit the plug gage within the "GO - NO GO" limits specified in 4.4.1, shall be considered not acceptable. In addition, when the hub is fitted onto the plug gage as specified in 4.4.1, there shall be no movement of the hub on the gage when the cannula is pressed lightly from side to side in all directions. The entire hub cavity wall, down to the entrance of the lumen, as well as the exterior of the hub, shall be well plated with nickel.

3.2.3 Stylet. Each needle shall be fitted with a stylet. The wire of the stylet shall be capable of easy insertion into, and withdrawal from, the needle. The wire shall be of adequate length to permit one end to be permanently secured into a knob, and the opposite end to be ground flush with the cutting edge of the cannula (i.e., the beveled point of the cannula). The wire shall be of sufficient diameter to prevent movement of the stylet which might permit the cut tip of the stylet to be bent while the stylet is in position in the needle (i.e., bevel of the stylet tip shall blend with bevel of point of the cannula presenting a single uniform surface with point of cannula). The knob shall be of conventional design having a taper tip that shall properly fit the taper slip of the needle hub, and a guide of adequate style and design to make a positive fit into the slotted edge of the needle hub and hold the ground edge of the wire stylet in proper position. The slot and guide shall be aligned with, and on the same side as the beveled surface of the cannula point.

3.3 Freedom from foreign matter. When tested as specified in 4.4.2, the hub cavity shall be substantially free from chips and other foreign matter. In addition, a visual examination of the lumen of the cannula shall reveal no presence of foreign matter, chips, burrs, or occlusion of the lumen.

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3.4 Security of assembly. When tested as specified in 4.4.3, the cannula shall not loosen or pull out of the hub.

3.5 Fluid tightness. When tested as specified in 4.4.4, the connection of the cannula with the hub shall not show leakage, and the connection of the hub with the test gage shall not show leakage exceeding the formation of a falling drop.

3.6 Identification marking.

3.6.1 Needles. The spinal puncture needles shall have the gage number and contractor's name or registered trademark die-stamped on the hubs.

3.6.2 Stylet knob and needle hub. The stylet knob and needle hub shall be permanently marked with a corresponding serial number (or serial number and letter). There shall be no duplication of serial numbers among needles of the same type and size delivered on any particular contract.

3.7 Workmanship. Workmanship shall be first class throughout. Needles shall be free from defects which detract from their appearance or may impair their serviceability. The cannula shall show a smooth finished outer surface free from pits and tool marks, burrs, scale or corrosion, and inner surface free from scale, deep pits, deep furrows, or die marks; the lumen of the cannula and the hub cavity shall be free from burrs, chips, dirt, slivers or other foreign matter. Needles shall be sharp, not blunt or bent, and points shall be free of feathered edges, burrs, or other defects, when examined under 7 power magnification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I.

Table I. Sampling Criteria for examination

Type defect	Inspection level	AQL (percent defective)
Major A	S4	1.0
Major B	S1	1.0
Minor	S4	6.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II.

Table II. Sampling Criteria for tests

Component	Characteristic	Reqmt.	Test procedure	Unit of product	Insp. level	AQL
Needle Cannula and Stylet Wire	Boil Test	GG-I-526	GG-I-526	1 needle	S2	*
Needle Hub and Stylet Knob	Plating Test	GG-I-526	QQ-N-290, or QQ-N-290 and QQ-C-320	1 needle	S2	*
Spinal puncture needle	Freedom from foreign matter	3.3	4.4.2	1 needle	S4	1.5
"	Security of assembly	3.4	4.4.3	1 needle	S3	1.5
"	Fluid tightness	3.5	4.4.4	1 needle	S4	1.5

* Acceptance number shall be zero for all sample sizes.

4.2.3 Raw materials. A sufficient quantity of each lot of metal used in the construction of the needles shall be selected for chemical analysis.

4.2.4 Test samples. Where possible, the same sample unit shall be used for determination of two or more test characteristics.

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4.3 Examination. The needles shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Inspection, however, shall not be restricted to defects listed in table III. Any needles in the sample containing one or more defects shall be rejected and, if the number of defective needles in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table III. Classification of defects in accordance with MIL-STD-105

Categories	Defects
Critical:	None defined
Major A	
101	Cannula lumen not free of obstructions.
102	Cannula lumen not free of scale, burrs, chips, dirt, deep pits, deep furrows, die marks, and other foreign material.
103	Cannula outer surface not smoothly finished and free of pits, tool marks, burrs, scale, and corrosion.
104	Cannula not free of cracks.
105	Cannula not concentric with hub.
106	Cannula outside diameter not as specified.
107	Point missing.
108	Point blunt or bent, not sharp, not symmetrical or properly formed, or not free of burrs, feathered edges, or other detrimental defects (examined under 7X magnification).
109	Point does not face flat of hub.
110	Hub not free of cracks, pits and tool marks.
111	Hub not free of burrs and slivers, and plating not free of flaking.
112	Hub interior not nickel plated down to entrance of lumen; exterior not nickel plated or chromium over nickel plated.
113	Hub cavity diameter and depth not in accordance with USA Standard as specified.
114	Hub indexing slot missing, or does not make proper fit with stylet lug.
115	Hub indexing slot not lined up with cannula point as specified.

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Major A:

116	Stylet missing.
117	Stylet tip not flush with cannula point.
118	Stylet outside diameter not as specified (fit).
119	Stylet lug missing, or not effective.
120	Stylet makes poor fit with needle.
121	Serial numbers (on hub and stylet) missing, incomplete, not legible or incorrect.
122	Serial number duplication (each duplication shall be counted as a defective unit).

Major B:

151	Cannula inside diameter not as specified.
152	Point length not as specified.
153	Cannula hole through wall at point not as specified.

Minor:

[illegible]	Discoloration of outside surfaces of needle or stylet.
202	Cannula length not as specified.
203	Identification marking missing, incomplete, not legible, or incorrect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Diameter and depth of Luer cavity. The Luer cavity shall be tested with the male Luer taper gage of figure 5 of USA Standard Z70.1. (This gage shall be as shown in the USA Standard, except that it may be constructed with a suitable fluid path to permit performance of the fluid tightness tests). The needle hub shall be slipped onto this gage with very light pressure. The fit of the needle hub on this gage shall be tested by means of the feeler gage in accordance with figures 5 and 6 of USA Standard Z70.1.

4.4.2 Freedom from foreign matter test. The Luer cavity shall be examined under 7X magnification, and good illumination, for chips and other foreign matter. In addition, a cotton swab, tightly wrapped on the end of a small wooden applicator, shall be vigorously turned down into the bottom of cavity, withdrawn, and examined for chips and other foreign matter under 7X magnification and good illumination. The cannula shall be tested for blockage by inserting a wire through the lumen of the cannula.

4.4.3 Security of assembly. A straight pull of 5 pounds shall be applied between the cannula and hub for a period of 1 minute.

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4.4.4 Fluid tightness test. With the needle in a horizontal position, needle hub shall be attached by means of a plug gage, as shown in figure 5 of USA Standard Z70.1-1955 (except that gage shall have been provided with a fluid path), to a water pressure source having a pressure gage. The water shall be permitted to run through the needle so as to fill the cavity space and the lumen of the cannula. The lumen shall then be sealed, and a pressure of 45 pounds per square inch gage (p.s.i.g) shall be applied for 10 seconds.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One needle, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

Table IV. Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	480 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive materials.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method 1A-15.

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5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States for America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Size (gage and length) required (see 1.2).

(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types etc., which are normally procured to meet Military requirements.

Custodians:
Army. - MD
Navy. - MS
Air Force .- 03

Preparing activity:
Defense Supply Agency. - DM

Review activities:
Army. - MD
Navy. - MS
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION: MIL-N-36220a

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 62 WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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